



HURDLE-STEP SCREEN

Objective: To examine simultaneous mobility of one limb and stability of the contralateral limb while maintaining both hip and torso stabilization during a balance challenge of standing on one leg

Equipment:

- Two uprights to anchor string (chair or table legs)
- 36-inch (0.9-m) piece of string
- 48-inch (1.2-m) wooden or plastic dowel

Instructions:

- Briefly discuss the protocol so the client understands what is required.
- Fasten a piece of string spanning two points at a height even with the underside of the foot positioned parallel with the floor, when it is raised to a height that flexes the hip to 70 degrees (approximately just above halfway up the tibia).
- Have the client stand with both feet together and the front edge of the toes aligned directly beneath the string.
 - ✓ Ultimately, this test should be performed with the feet positioned at gait-width apart [i.e., 2.8 to 3.5 inches (7 to 9 cm)] to simulate single-leg support during walking.
- Place the dowel across the client's shoulders, holding it parallel to the floor (similar to the placement of the bar during the traditional barbell squat).
- Instruct the client to load onto one leg and slowly lift the opposite leg over the string, flexing the hip to clear the string and then gently touching the heel of the raised leg to the floor in front of the string before returning to the starting position.
 - ✓ The foot only needs to clear the string and does not need to be lifted as high as possible.
 - ✓ It is important to remember not to cue the client to use good technique, instead observing the natural movement.
 - ✓ Repetitions need to be performed slowly and with control.
- Have the client repeat the movement with the opposite leg, completing a series of repetitions with each leg so that the CMES can make the necessary evaluations.
- Allow sufficient practice trials to accommodate learning before administering the test screens.

Observations (see Table on the next page):

- Frontal view (Figure 1):
 - ✓ First repetition: Observe the stability of the foot (i.e., evidence of pronation, supination, eversion, or inversion).
 - ✓ Second repetition: Observe the alignment of the stance-leg knee over the foot (i.e., evidence of knee movement in any plane).
 - ✓ Third repetition: Watch for excessive hip adduction greater than 2 inches (5.1 cm), as measured by excessive stance-leg adduction or downward hip-tilting toward the opposite side (Figure 2).
 - ✓ Fourth repetition: Observe the stability of the torso (i.e., evidence of torso movement in any plane as demonstrated by movement of the dowel) (see Figure 2).
 - ✓ Fifth repetition: Observe the alignment of the moving leg (i.e., lack of dorsiflexion at the ankle, deviation from the sagittal plane at the knee or ankle, or hiking of the moving hip) (see Figure 2).
- Sagittal view (Figure 3):
 - ✓ First repetition: Observe the stability of the torso and stance leg.
 - ✓ Second repetition: Observe the mobility of the hip (i.e., allowing 70 degrees of hip flexion without compensation—anterior tilting).

General interpretations:

- Identify the origin(s) of movement limitation or compensation.
- Evaluate the impact on the entire kinetic chain.

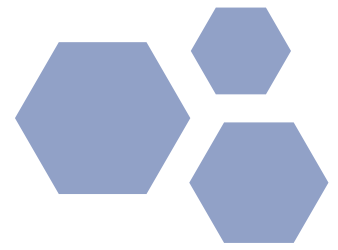


Figure 1
Hurdle-step screen: Anterior view



Figure 2
Hurdle-step screen: Anterior view with compensations



Figure 3
Hurdle-step screen: Sagittal view

HURDLE-STEP SCREEN

View		Joint Location	Compensation	Key Suspected Compensations: Overactive (Tight)	Key Suspected Compensations: Underactive (Lengthened)
☐	From the front	Feet	Lack of foot stability: Ankles collapse inward/ feet turn outward	Soleus, lateral gastrocnemius, peroneals	Medial gastrocnemius, gracilis, sartorius, tibialis group, gluteus medius and maximus—inability to control internal rotation
☐	From the front	Knees	Move inward	Hip adductors, tensor fascia latae	Gluteus medius and maximus
☐	From the front	Hips	Hip adduction* >2 inches (5.1 cm) Stance-leg hip rotation (inward)	Hip adductors, tensor fascia latae Stance-leg or raised-leg internal rotators	Gluteus medius and maximus Stance-leg or raised-leg external rotators
☐	From the front	Torso	Lateral tilt, forward lean, rotation	Lack of core stability	
☐	From the front	Raised-leg	Lack of ankle dorsiflexion Limb deviates from sagittal plane Hiking the raised hip	Ankle plantarflexors Raised-leg hip extensors Stance-leg hip flexors—limiting posterior hip rotation during raise	Ankle dorsiflexors Raised-leg hip flexors
☐	From the side	Pelvis and low back	Anterior tilt with forward torso lean Posterior tilt with hunched-over torso	Stance-leg hip flexors Rectus abdominis and hip extensors	Rectus abdominis and hip extensors Stance-leg hip flexors

*Hip adduction involves weight transference over the stance leg while preserving hip, knee, and foot alignment. This weight transference requires a 1- to 2-inch (2.5- to 5-cm) lateral shift over the stance-leg, with a small hike in the stance-hip of 4 to 5 degrees or less.

Sources: Kendall, F.P. et al. (2005). *Muscles Testing and Function with Posture and Pain* (5th ed.). Baltimore, Md.: Lippincott Williams & Wilkins; Cook, G. (2003). *Athletic Body in Balance*. Champaign, Ill.: Human Kinetics; Sahrmann, S.A. (2002). *Diagnosis and Treatment of Movement Impairment Syndromes*. St. Louis, Mo.: Mosby.